Siddharth Dongardive

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SUMMARY

Data Science undergraduate who builds end-to-end, leakage-free analytics and machine learning projects on real-world data using Python, SQL, PostgreSQL, XGBoost, and PyTorch. Experienced in time-series forecasting, causal analysis, cost-sensitive classification, temporal model evaluation, and drift diagnosis. Focused on rigorous evaluation and on honestly reporting what works and what does not.

EDUCATION

Vidyavardhini's College of Engineering and Technology, Mumbai 2023 – 2027
B.E. Computer Science and Engineering (Data Science) **CGPA: 7.5 / 10**

SKILLS

Programming:	Python, SQL, C++
Data Science and Statistics	Pandas, NumPy, Scikit-learn, Feature Engineering, EDA, Time-Series Forecasting, Causal Inference (Interrupted Time Series, Placebo Tests), Cost-Sensitive Classification
Machine Learning:	XGBoost, PyTorch, PyTorch Geometric (GraphSAGE, RGCN, HGT), SHAP, Imbalanced Learning, Temporal Validation, Drift Detection
LLM and Retrieval:	RAG, FAISS, BM25, Pinecone, Cross-Encoder Reranking, LLM-as-Judge Evaluation, CrewAI
Databases and ETL:	PostgreSQL, MongoDB, ETL, Data Cleaning, REST APIs
Tools:	FastAPI, Streamlit, Docker, Git, GitHub, Power BI, MLflow, DVC

PROJECTS

Delhi Air Quality Risk Forecasting | Python, PostgreSQL, XGBoost, Scikit-learn, Streamlit
GitHub | Live dashboard | Kaggle dataset

- Built an end-to-end pipeline from raw CPCB/CAAQMS data (7 Delhi stations, 2023–2026) into PostgreSQL with 124 engineered features for 1h, 6h, and 24h forecasting, improving 24h MAE to 34.11 AQI from 38.34 for naive persistence.
- Designed cost-aware alert thresholds: at a 5:1 missed-hazard to false-alarm cost, Severe-AQI recall rose from 61.1% to 95.7% (precision 73.9% to 54.3%) versus the fixed CPCB cutoff.
- Ran a causal analysis of the Diwali firecracker ban (interrupted time series, fixed effects, placebo test): a naive +135.6 AQI effect became statistically zero (-10.6, p=0.43), and the placebo showed a spurious +24.3 from seasonal confounding.

BitcoinGraphGuard: Temporal Graph Fraud Detection | Python, PyTorch Geometric, XGBoost, SHAP, FastAPI, Docker
GitHub | Live demo

- Built a leakage-free temporal evaluation pipeline on the Elliptic++ dataset (203,769 transactions, 822,942 wallets, 4.4M edges, 49 time steps). Corrected 347,569 duplicate wallet rows before modeling.
- Benchmarked XGBoost (0.801 PR-AUC), GraphSAGE (0.621), HGT (0.486), and RGCN (0.468) under one strict temporal split at 6.5% illicit prevalence; a GraphSAGE vs MLP ablation showed graph structure adds +30% PR-AUC.
- Diagnosed the collapse on steps 43–49 (PR-AUC about 0.04–0.05) as covariate shift (adversarial validation AUC 1.0, no timestamp leaks) and explained predictions with SHAP (Local_feature_53 dominant).

LegalRAG: Hybrid Retrieval over Indian High Court Judgments | Python, FAISS, BM25, Sentence-Transformers, FastAPI
Live demo | GitHub

- Built a hybrid retrieval pipeline over 100,000 High Court judgments (538,079 chunks) combining BM25, dense retrieval (bge-base, FAISS), Reciprocal Rank Fusion, and cross-encoder reranking.
- Created a 497-question benchmark across 5 question types (fact, reasoning, legal provision, outcome, multi-hop) with an LLM-judge harness scoring faithfulness, citation correctness, and hallucination rate.
- Found the reranker improved Recall@1 but slightly hurt Recall@5/10 because of domain shift on legal text, and packaged the system as a tested Python module (133 pytest tests) with a live demo.

TruthMates: Civic Misinformation Detection | Python, CrewAI, Pinecone, MongoDB, FastAPI, React
GitHub | Demo available on request

- Designed and built a staged CrewAI pipeline (civic classification, evidence retrieval, counter-statement generation, validation, monitoring) behind a FastAPI API, persisting each stage's output to MongoDB.
- Retrieved evidence from a PIB-verified fact corpus in Pinecone and the Google Fact Check API, using an embedding-based civic classifier and tuned similarity thresholds.
- Added video and audio claim analysis (yt-dlp, Groq Whisper) and English/Hindi counter-statements with a trust score, served to a React frontend.

ACHIEVEMENTS

- **1st Place** — Line Following Robot, VIT (2026)
- **1st Place** — HackX Hackathon, SFIT (2026)

ACTIVITIES

- **Manager, Yantrika (VCET)** (Oct 2025 – Jul 2026) — team placed 3rd in Line Follower, Technoxian Delhi 2025.